

Prashanth Durgam

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PROFESSIONAL SUMMARY

Aspiring Software Engineer with hands-on experience in full-stack development using MERN, Next.js and Python. Passionate about building scalable, reliable applications and integrating AI-driven solutions to solve practical real-world problems.

EDUCATION

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|---|----------------------------------|
| • National Institute of Technology, Durgapur
<i>Master of Technology in Computer Science and Engineering</i> | 2024 – 2026
CGPA: 9.00/10 |
| • Chaitanya Bharathi Institute of Technology, Hyderabad
<i>Bachelor of Engineering in Electronics and Communication Engineering</i> | 2019 – 2023
CGPA: 8.09/10 |
| • Sri Chaitanya Junior Kalasala, Hyderabad
<i>Telangana State Board of Intermediate Education</i> | 2017 – 2019
Percentage: 95.9% |

PROJECTS

- **Automated Personal Comfort Recommendation System | Generative AI Model [GitHub]**
M.Tech Major Project
 - **Tools & Technologies:** ReactJS, ExpressJS, Python, FastAPI, Gemini 2.5-flash, and Pydantic.
 - Developed a full-stack AI system integrating a comfort recommendation engine with an interactive Gemini-powered conversational assistant.
 - Engineered dynamic appliance control heuristics using real-time indoor, outdoor, and environmental sensor telemetry.
 - Implemented a user feedback loop and schema validation using Pydantic to enhance recommendation precision and response reliability.
- **Document Q&A Chatbot | Retrieval-Augmented Generation (RAG) [GitHub]**
Generative AI Application
 - **Tools & Technologies:** Python, LangChain, Google Gemini, Streamlit, PyPDF, Vector Embeddings.
 - Built an end-to-end RAG chatbot enabling real-time, context-grounded Q&A on user-uploaded PDF documents.
 - Implemented document chunking, semantic vector indexing, and similarity search using Google Gemini Embeddings and LangChain.
 - Developed a responsive Streamlit chat interface with multi-turn conversation memory and dynamic prompt orchestration.

PUBLICATIONS

- **AirWise: LLM-Powered Smart Air Quality Monitoring and Automated Appliance Control, AIEI 2026.**
- **iAirGuard: A Modular IoT Architecture with Dynamic Sampling and TinyML-Based Fault Detection for Air Quality Monitoring, IEEE COMSNETS 2026.**

TECHNICAL SKILLS AND INTERESTS

Languages: Java, Python, C, JavaScript
Web Tech: HTML, CSS, React.js, Node.js, Express.js, FastAPI, REST APIs
AI & DB: LLM, LangChain, RAG, Agentic AI, SQL, MongoDB
Tools & Core: Git, GitHub, VS Code, Data Structures & Algorithms, OOP
Spoken: English, Hindi, Marathi, Telugu

CERTIFICATES & ACHIEVEMENTS

- **Qualified GATE 2024**
- **Python Programming:** Complete Python Bootcamp (Udemy)
- **SQL Fundamentals:** SQL Certification (CodeChef)